

Spheroidal Wave Functions

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1 Background: Spheroidal Wave Functions

This presentation follows Flammer [1]. Recall's Yukawa equation,

$$\Delta\psi - \lambda^2\psi = 0$$

Re-writing the above explicitly in prolate and oblate spheroidal coordinates, respectively, yields

$$\left[\frac{\partial}{\partial v} (1 - v^2) \frac{\partial}{\partial v} + \frac{\partial}{\partial u} (u^2 - 1) \frac{\partial}{\partial u} + \frac{u^2 - v^2}{(u^2 - 1)(1 - v^2)} \frac{\partial^2}{\partial \phi^2} - a^2 \lambda^2 (u^2 - v^2) \right] \psi = 0.$$

and

$$\left[\frac{\partial}{\partial v} (1 - v^2) \frac{\partial}{\partial v} + \frac{\partial}{\partial u} (u^2 + 1) \frac{\partial}{\partial u} + \frac{u^2 + v^2}{(u^2 + 1)(1 - v^2)} \frac{\partial^2}{\partial \phi^2} - a^2 \lambda^2 (u^2 + v^2) \right] \psi = 0.$$

A standard separation-of-variables argument with $\psi = U(u)V(v)e^{im\phi}$ yields the following ODEs,

$$\begin{aligned} \text{prolate} &= \begin{cases} \frac{d}{du} \left((u^2 - 1) \frac{dU}{du} \right) - \left[\Lambda + a^2 \lambda^2 u^2 + \frac{m^2}{u^2 - 1} \right] U = 0 \\ \frac{d}{dv} \left((1 - v^2) \frac{dV}{dv} \right) + \left[\Lambda + a^2 \lambda^2 v^2 - \frac{m^2}{1 - v^2} \right] V = 0 \end{cases} \\ \text{oblate} &= \begin{cases} \frac{d}{du} \left((u^2 + 1) \frac{dU}{du} \right) - \left[\Lambda - a^2 \lambda^2 u^2 + \frac{m^2}{u^2 + 1} \right] U = 0 \\ \frac{d}{dv} \left((1 - v^2) \frac{dV}{dv} \right) + \left[\Lambda + a^2 \lambda^2 v^2 - \frac{m^2}{1 - v^2} \right] V = 0 \end{cases} \end{aligned}$$

Let's standardize our notation with the substitution $c = \lambda a$ to yield

$$\begin{aligned} \text{prolate} &= \begin{cases} \frac{d}{du} \left((u^2 - 1) \frac{dU}{du} \right) - \left[\Lambda + c^2 u^2 + \frac{m^2}{u^2 - 1} \right] U = 0 \\ \frac{d}{dv} \left((1 - v^2) \frac{dV}{dv} \right) + \left[\Lambda + c^2 v^2 - \frac{m^2}{1 - v^2} \right] V = 0 \end{cases} \\ \text{oblate} &= \begin{cases} \frac{d}{du} \left((u^2 + 1) \frac{dU}{du} \right) - \left[\Lambda - c^2 u^2 + \frac{m^2}{u^2 + 1} \right] U = 0 \\ \frac{d}{dv} \left((1 - v^2) \frac{dV}{dv} \right) + \left[\Lambda + c^2 v^2 - \frac{m^2}{1 - v^2} \right] V = 0 \end{cases} \end{aligned}$$

In accordance with [1], we denote $U(u) = R_{nm}(u; ic)$ and $V(v) = S_{nm}(v; ic)$ as the prolate radial and angular spheroidal wave functions, respectively. For the oblate case, we write $U(u) = R_{nm}(iu; c)$ and $V(v) = S_{nm}(u; c)$. Note that in both of the prolate and oblate cases, R_{nm} and S_{nm} refer to the same functions.

We remark that there are multiple solutions for the angular and radial ODEs. For the angular ODE, we shall only use the angular function of the first kind, denoted $S_{nm}(u; c)$. For the radial ODE, we use only the radial function of the first and third kind, which is denoted $R_{nm}^{(1)}(u; c)$ and $R_{nm}^{(3)}(u; c)$, respectively.

As is the case with other PDEs, we are able to expand out solutions φ to the Yukawa BVP in the interior and exterior of the spheroid, as follows,

$$\varphi_{\text{prolate}}(u, \theta, \phi) = \begin{cases} \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm}^o R_{nm}^{(3)}(u; ic) S_{nm}(v; ic) e^{im\phi} & u > u_0 \\ \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm}^i R_{nm}^{(1)}(u; ic) S_{nm}(v; ic) e^{im\phi} & 1 < u < u_0, \end{cases}$$

and

$$\varphi_{\text{oblate}}(r, \theta, \phi) = \begin{cases} \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm}^o R_{nm}^{(3)}(iu; c) S_{nm}(v; c) e^{im\phi} & u > u_0 \\ \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm}^i R_{nm}^{(1)}(iu; c) S_{nm}(v; c) e^{im\phi} & 0 < u < u_0, \end{cases}$$

The Wronskian relationship of the radial function of the first and third kind will also be used [1], and they are given as, in prolate and oblate coordinates, respectively,

$$\begin{aligned} W(R_{nm}^{(1)}, R_{nm}^{(3)}) &= iW(R_{nm}^{(1)}, R_{nm}^{(2)}) \\ &= \frac{i}{c(u_0^2 - 1)} \end{aligned}$$

and

$$\begin{aligned} W(R_{nm}^{(1)}, R_{nm}^{(3)}) &= iW(R_{nm}^{(1)}, R_{nm}^{(2)}) \\ &= \frac{i}{c(u_0^2 + 1)} \end{aligned}$$

2 Spectra for Modified Layer Potentials

2.1 Single-Layer Potential

We only derive out the prolate case; the process for oblate is analogous. Note that any prolate formula can be quickly turned into an oblate formula by the substitution $u \mapsto iu$ and $c \mapsto -ic$.

Consider the ansatz $\tilde{\sigma} = \frac{1}{\sqrt{u_0^2 - v^2}} S_{nm}(u_0; ic) e^{im\phi}$. Let $\varphi = S_\lambda[\tilde{\sigma}]$. Then, we have the following expansion,

$$\varphi(u, \theta, \phi) = \begin{cases} \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm}^o R_{nm}^{(3)}(u; ic) S_{nm}(v; ic) e^{im\phi} & u > u_0 \\ \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm}^i R_{nm}^{(1)}(u; ic) S_{nm}(v; ic) e^{im\phi} & 1 < u < u_0, \end{cases}$$

where a_{nm}^o and a_{nm}^i represents the exterior and interior coefficients, respectively. Now, we use the orthogonality of $\{S_{nm}(v; ic)e^{im\phi}\}$ and use the standard jump relations $[[S_\lambda \tilde{\sigma}]] = 0$ and $[[S'_\lambda \tilde{\sigma}]] = -\tilde{\sigma}$ to obtain

$$a_{nm}^o R_{nm}^{(3)}(u_0; ic) - a_{nm}^i R_{nm}^{(1)}(u_0; ic) = 0$$

$$\frac{\sqrt{u_0^2 - 1}}{a} (a_{nm}^o [\partial_u R_{nm}^{(3)}(u_0; ic)] - a_{nm}^i [\partial_u R_{nm}^{(1)}(u_0; ic)]) = -1$$

Solving the 2×2 system for a_{nm}^o and a_{nm}^i , we get

$$a_{nm}^o = iac\sqrt{u_0^2 - 1} R_{nm}^{(1)}(u_0; ic)$$

$$a_{nm}^i = iac\sqrt{u_0^2 - 1} R_{nm}^{(3)}(u_0; ic)$$

Thus,

$$S_\lambda^{\text{pro}}[\tilde{\sigma}] = \begin{cases} iac\sqrt{u_0^2 - 1} R_{nm}^{(3)}(u_0; ic) R_{nm}^{(1)}(u; ic) S_{nm}(v; ic) e^{im\phi}, & 1 < u < u_0, \\ iac\sqrt{u_0^2 - 1} R_{nm}^{(1)}(u_0; ic) R_{nm}^{(3)}(u; ic) S_{nm}(v; ic) e^{im\phi}, & u > u_0. \end{cases}$$

Taking the normal derivative, we yield

$$\partial_n S_\lambda^{\text{pro}}[\tilde{\sigma}] = \begin{cases} ic\sqrt{u_0^2 - 1} \sqrt{\frac{u^2 - 1}{u^2 - v^2}} R_{nm}^{(3)}(u_0; ic) \partial_u R_{nm}^{(1)}(u; ic) Y_{nm}(v, \phi), & 1 < u < u_0, \\ ic\sqrt{u_0^2 - 1} \sqrt{\frac{u^2 - 1}{u^2 - v^2}} R_{nm}^{(1)}(u_0; ic) \partial_u R_{nm}^{(3)}(u; ic) Y_{nm}(v, \phi), & u > u_0. \end{cases}$$

Similarly, for the oblate case with the ansatz $\tilde{\sigma} = \frac{1}{\sqrt{u_0^2 + v^2}} S_n^m(v; c) e^{im\phi}$, we have

$$S_\lambda^{\text{obl}}[\tilde{\sigma}] = \begin{cases} -ac\sqrt{u_0^2 + 1} R_{nm}^{(3)}(iu_0; c) R_{nm}^{(1)}(iu; c) S_{nm}^{(1)}(v; c) e^{im\phi}, & 0 < u < u_0, \\ -ac\sqrt{u_0^2 + 1} R_{nm}^{(1)}(iu_0; c) R_{nm}^{(3)}(iu; c) S_{nm}^{(1)}(v; c) e^{im\phi}, & u > u_0. \end{cases}$$

and

$$\partial_n S_\lambda^{\text{obl}}[\tilde{\sigma}] = \begin{cases} -c\sqrt{u_0^2 + 1} \sqrt{\frac{u^2 + 1}{u^2 + v^2}} R_{nm}^{(3)}(iu_0; c) R_{nm}'^{(1)}(iu; c) S_{nm}(v; c), & 0 < u < u_0, \\ -c\sqrt{u_0^2 + 1} \sqrt{\frac{u^2 + 1}{u^2 + v^2}} R_{nm}^{(1)}(iu_0; c) R_{nm}'^{(3)}(iu; c) S_{nm}(v; c), & u > u_0. \end{cases}$$

2.2 Double-Layer Potential

We only prove the prolate case here. Let's use the ansatz $\tilde{\sigma} = S_{nm}(u_0; ic)e^{im\phi}$. Then, we have the following expansion,

$$\varphi(u, \theta, \phi) = \begin{cases} \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm}^o R_{nm}^{(3)}(u; ic) S_{nm}(v; ic) e^{im\phi} & u > u_0 \\ \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm}^i R_{nm}^{(1)}(u; ic) S_{nm}(v; ic) e^{im\phi} & 1 < u < u_0, \end{cases}$$

Consider the angular ansatz $\tilde{\sigma} = S_{nm}(u_0; ic)$. Let $\varphi(u, v, \phi) = D_\lambda[\tilde{\sigma}](u, v, \phi)$. Then, we have that

$$\varphi(r, \theta, \phi) = \begin{cases} \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm}^o R_{nm}^{(3)}(u; ic) S_{nm}(v; ic) e^{im\phi} & u > u_0 \\ \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm}^i R_{nm}^{(1)}(u; ic) S_{nm}(v; ic) e^{im\phi} & 1 < u < u_0, \end{cases}$$

where a_{nm}^o and a_{nm}^i represents the exterior and interior coefficients, respectively. Now, we use the orthogonality of $\{S_{nm}(v; ic)e^{im\phi}\}$ and use the jump relations $[[DT_n^m]] = T_n^m$ and $[[D'T_n^m]] = 0$ to yield the following system,

$$\begin{aligned} a_{nm}^o R_{nm}^{(3)}(u_0; ic) - a_{nm}^i R_{nm}^{(1)}(u_0; ic) &= 1 \\ a_{nm}^o [\partial_u R_{nm}^{(3)}(u_0; ic)] - a_{nm}^i [\partial_u R_{nm}^{(1)}(u_0; ic)] &= 0 \end{aligned}$$

Solving this 2×2 system yields the following solutions for the exterior and interior coefficients,

$$\begin{aligned} a_{nm}^o &= -\frac{\partial_u R_{nm}^{(1)}(u_0; ic)}{W(R_{nm}^{(1)}, R_{nm}^{(3)})} \\ &= ic(u_0^2 - 1) \partial_u R_{nm}^{(1)}(u_0; c) \\ a_{nm}^i &= -\frac{\partial_u R_{nm}^{(3)}(u_0; ic)}{W(R_{nm}^{(1)}, R_{nm}^{(3)})} \\ &= ic(u_0^2 - 1) \partial_u R_{nm}^{(3)}(u_0; c) \end{aligned}$$

Thus,

$$D_\lambda^{\text{pro}}[\tilde{\sigma}] = \begin{cases} \left[ic(u_0^2 - 1) \partial_u R_{nm}^{(3)}(u_0; ic) \right] R_{nm}^{(1)}(u; ic) S_{nm}(v; ic) e^{im\phi}, & 1 \leq u < u_0 \\ \left[ic(u_0^2 - 1) \partial_u R_{nm}^{(1)}(u_0; ic) \right] R_{nm}^{(3)}(u; ic) S_{nm}(v; ic) e^{im\phi}, & u > u_0 \end{cases}$$

Taking the normal derivative yields

$$\partial_n D_\lambda^{\text{pro}}[\tilde{\sigma}] = \begin{cases} \frac{ic(u_0^2 - 1)}{a} \sqrt{\frac{u^2 - 1}{u^2 - v^2}} [\partial_u R_{nm}^{(3)}(u_0; ic)] [\partial_u R_{nm}^{(1)}(u; ic)] S_{nm}(v; ic) e^{im\phi}, & u < u_0, \\ \frac{ic(u_0^2 - 1)}{a} \sqrt{\frac{u^2 - 1}{u^2 - v^2}} [\partial_u R_{nm}^{(1)}(u_0; ic)] [\partial_u R_{nm}^{(3)}(u; ic)] S_{nm}(v; ic) e^{im\phi}, & u > u_0. \end{cases}$$

Similarly, for the oblate case, we have the following,

$$D_{\lambda}^{\text{obl}} [\tilde{\sigma}] = \begin{cases} -c (u_0^2 + 1) R_{nm}^{(3)'} (iu_0; c) R_{nm}^{(1)} (iu; c) S_{nm}^{(1)} (v; c) e^{im\phi}, & 0 < u < u_0, \\ -c (u_0^2 + 1) R_{nm}^{(1)'} (iu_0; c) R_{nm}^{(3)} (iu; c) S_{nm}^{(1)} (v; c) e^{im\phi}, & u > u_0, \end{cases}$$

and

$$\partial_n D_{\lambda} [\tilde{\sigma}] = \begin{cases} -\frac{c (u_0^2 + 1)}{a} \sqrt{\frac{u^2 + 1}{u^2 + v^2}} R_{nm}^{(3)'} (iu_0; c) R_{nm}^{(1)'} (iu; c) S_{nm} (v; c) e^{im\phi}, & 0 < u < u_0, \\ -\frac{c (u_0^2 + 1)}{a} \sqrt{\frac{u^2 + 1}{u^2 + v^2}} R_{nm}^{(1)'} (iu_0; c) R_{nm}^{(3)'} (iu; c) S_{nm} (v; c) e^{im\phi}, & u > u_0. \end{cases}$$

References

- [1] Carson Flammer. *Spheroidal Wave Functions*. eng. Dover Books on Mathematics. Newburyport: Dover Publications, 2014. ISBN: 978-0-486-44639-4.